Palo Alto Networks Cybersecurity Academy –Setting up Web URL filtering on a PA220 firewall

Sai Srinivas Pasyavula

Purpose

The purpose of this lab is to set up filtering of Websites and software on a PA220 firewall for an elementary school. It covers skills of creating a URL-filtering security profile, A SSL/TLS service profile, and certificate and a URL admin Override.

Background

Palo Alto Networks is a networking and cybersecurity company from Santa Clara, California. They are a member of the S&P 500. They focus mainly on the business market, creating scalable security solutions for many of the largest companies Worldwide.

The Palo Alto PA220 is a firewall sold by Palo Alto Networks. Contrary to Palo Alto's main market, the PA220 is intended for small office/home office solutions. The PA220 can also be used in a small school environment, as is the case in this lab. Marketed as a NGFW, or Next-Generation Firewall, the PA220 uses machine learning to identify attacks instead of relying on a simple signature check like traditional firewalls. This technology allows the PA220 to identify undocumented threats and brand-new exploits without intervention from Palo Alto networks themselves. As of January 31, 2023, it is no longer being sold, and it will reach end-of-life on January 31, 2028.

The latest version of PAN-OS is PAN-OS 11.2 Quasar, which was released in May 2024. In this lab, our firewall is running PAN-OS 8, which has reached end of life and is no longer supported.

PAN-OS's GUI has a variety of settings and tools to control advanced functionality of the router. The GUI's default page is a dashboard that displays vital information, such as console messages and link states of ports.

The PA220 is primarily intended for SOHO, or Small Office/Home office networks, which normally have ten or fewer employees and are perfect for small businesses. However, the PA220 also works in small-scale school environments, such as elementary school computer labs. This is the use case uponon which this lab is built.

The PA220 offers a variety of methods to filter and manage inbound and outbound traffic. One such method is URL filtering. A Uniform Resource Locator, or URL, is the primary identifier for hosts on the internet and can be used with a variety of protocols. In this case, the PA220 uses URLs to identify the destination of HTTP and HTTPS traffic.

Another filtering method offered by the PA220 is DNS-based filtering. Domain Name System, or DNS, is a method of mapping user-readable hostnames to computerreadable IP addresses. DNS uses port 53 and is unique in that it can use both TCP and UDP for data transmission. DNS works by having a central server keep a database of IP-hostname mappings and provide them to clients upon requests. The PA220 can filter through DNS by intercepting these requests and choosing to respond with either a fake sinkhole address or no address at all.

Yet another filtering method offered by the PA220 is application-level filtering. At layer 4 of the OSI model, communication is primarily managed by the Transmission Control Protocol and the User Datagram Protocol, both of which communicate different types of traffic by assigning port numbers to different application layer protocols. Common port number examples are 80 for HTTP, 22 for SSH, and 443 for HTTPS. The PA220 can easily filter this traffic by checking for specified port numbers. Applicationlevel filtering isn't as useful as DNS or URL filtering, as many protocols (DNS, HTTP(S), and DHCP) must be enabled for a network to function properly and most modern programs use HTTP(S) for any network functionality.

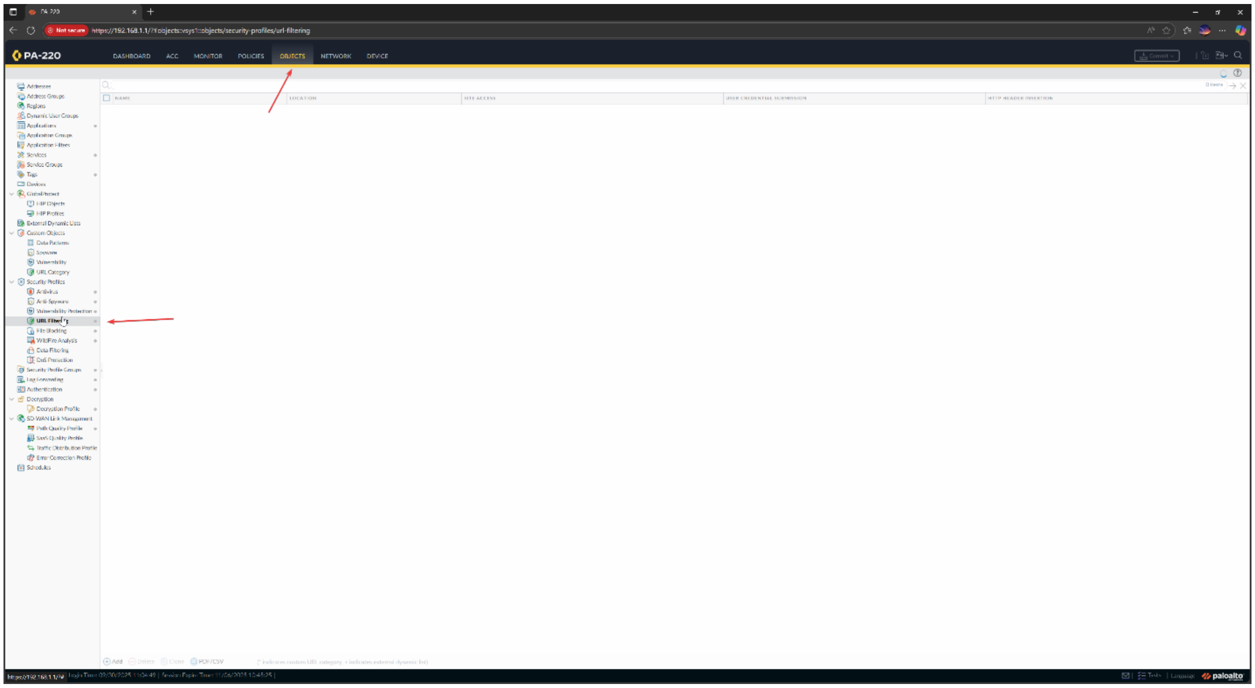
Another example of an application with its own port number is IRC. Internet Relay Chat, more commonly known as IRC, is an instant messaging protocol created in 1988 and popularized in the mid to late 1990s. It was created in Finland at the Universituy of Oulu and spread through various universities before eventually becoming popular with the general public. Nowadays, the protocol isn't commonly used for messaging, being replaced with web-based chat apps like Discord, Slack, and Microsoft Teams. However, IRC maintains a small cult following and has several active servers that keep it alive.

Lab Summary

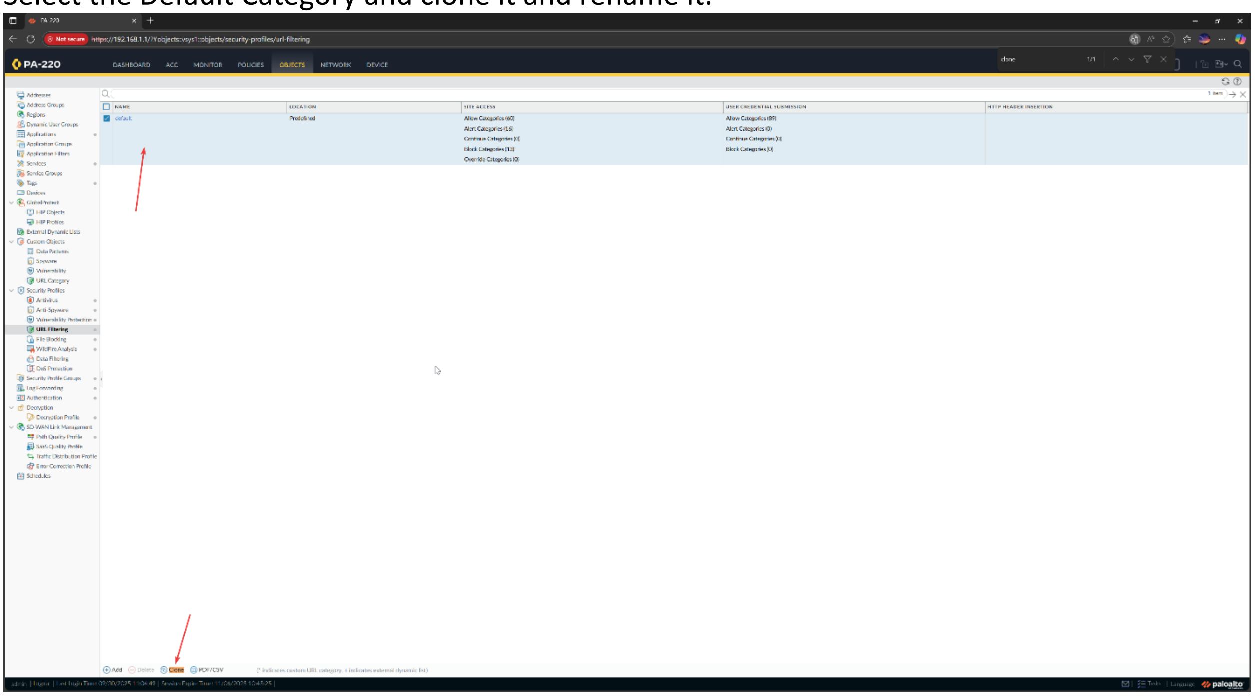
In this lab, we set up three two types of web filtering: URL filtering and a URL admin override. For URL filtering, we blocked all predefined categories that weren't appropriate for a Elementary school environment and created a custom URL category that was blocked by default but could be overridden with a password. We created a SSL/TLS profile to let the firewall present a valid HTTPS certificate when showing URL-filtering override/warning pages. Without it, the firewall can't display the override page for encrypted (HTTPS) sites.

Lab Commands

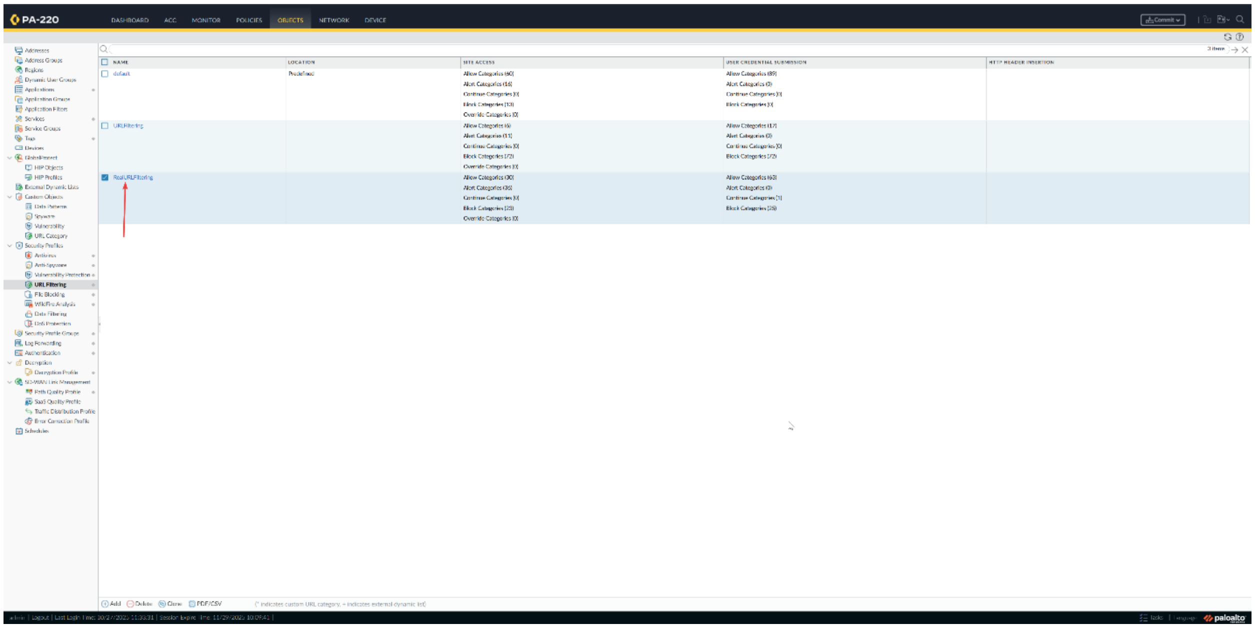
URL Filtering:
Go to Objects > Custom Objects > URL Category.



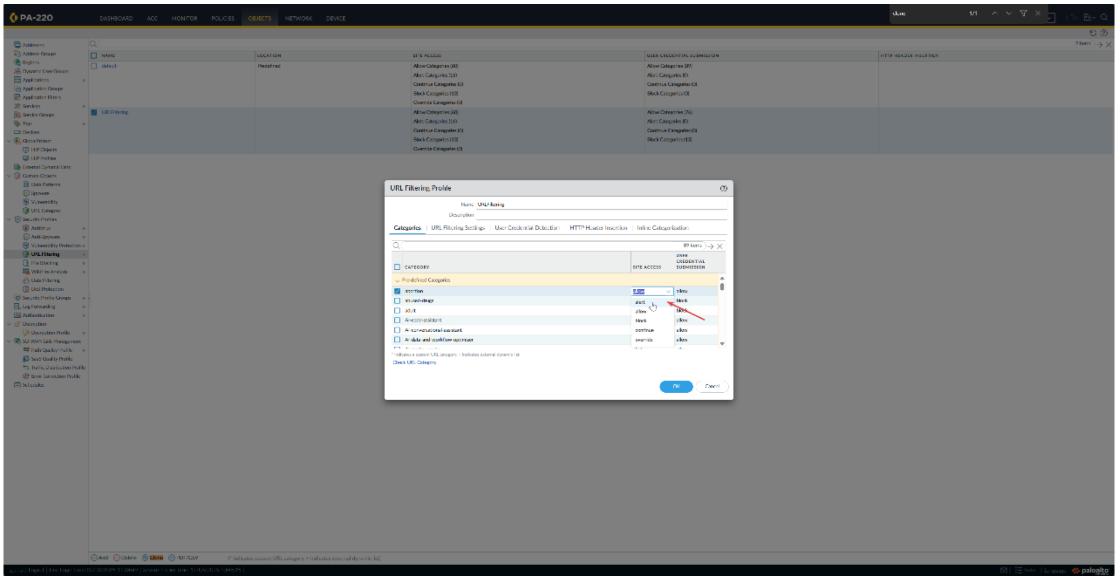
Select the Default Category and clone it and rename it.



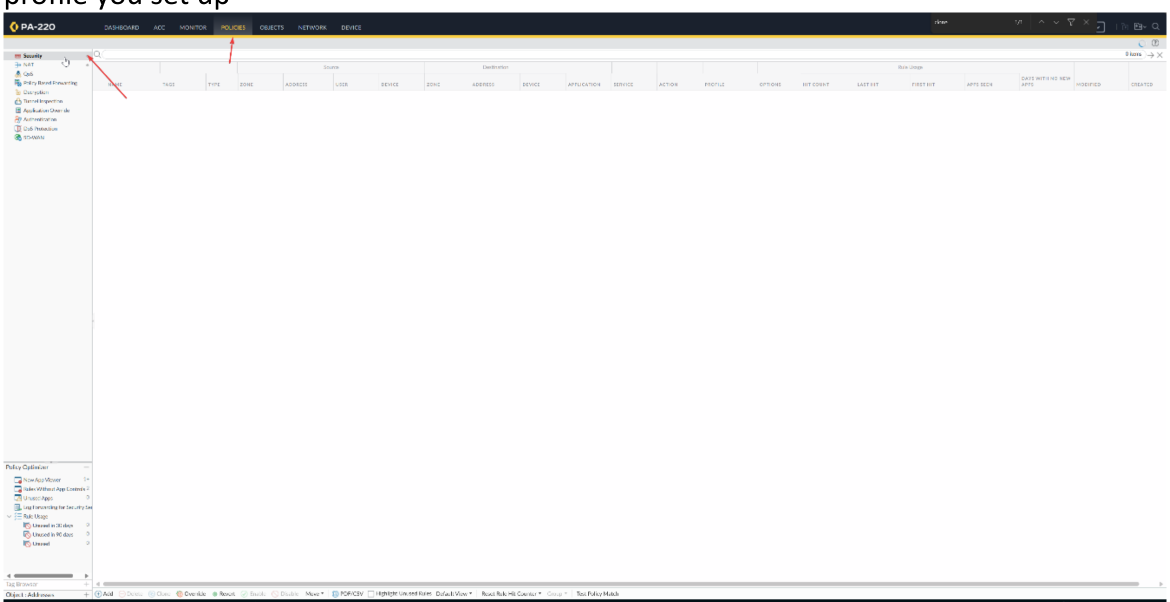
Name your profile and give it a description. Click the check next to your custom URL. For this lab I named my category RealURLFiltering. Click into it

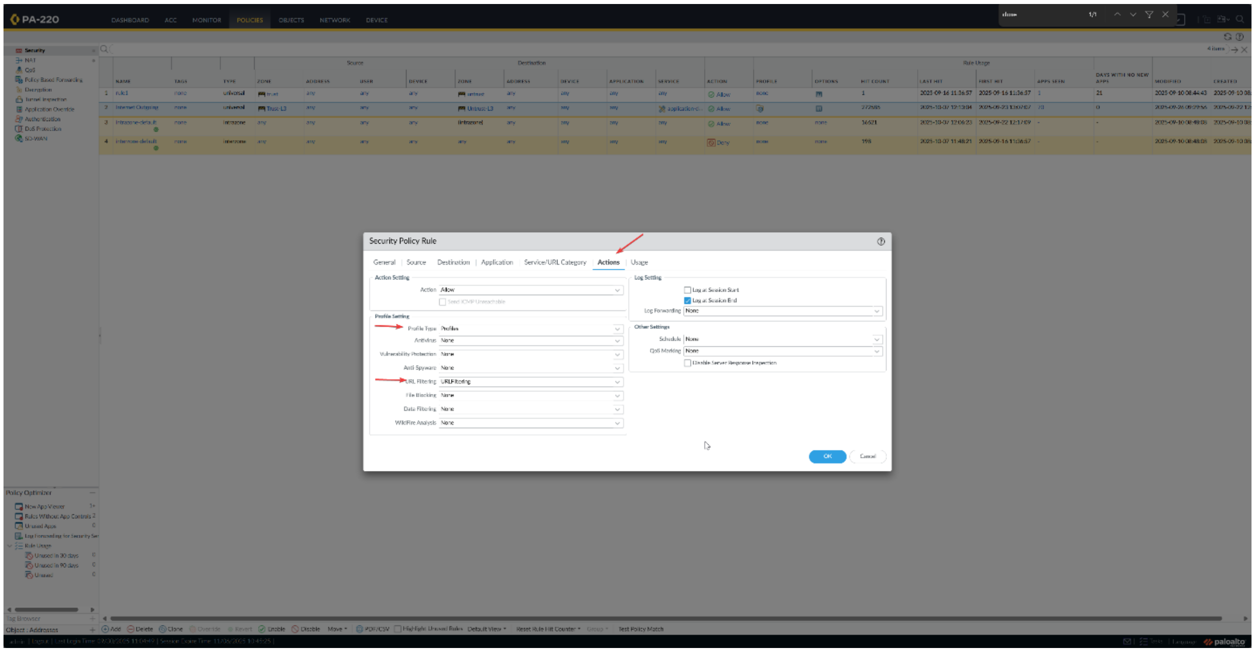


In the section that lists all URL categories, select all categories and then de-select malware, command-and-control, and phishing.
To the right of the Action column heading, mouse over and select the down arrow and then select Set Selected Actions and choose alert. Choose the appropriate action to match expectations for an Elementary School. Make sure “Site Access” and “User Credential Submission” are set to “block”. Optionally, you can set the “Site Access” key to “override”, which will make the website require an administrator password to access. Properly configure the pre-defined categories to match what would be appropriate for a school setting



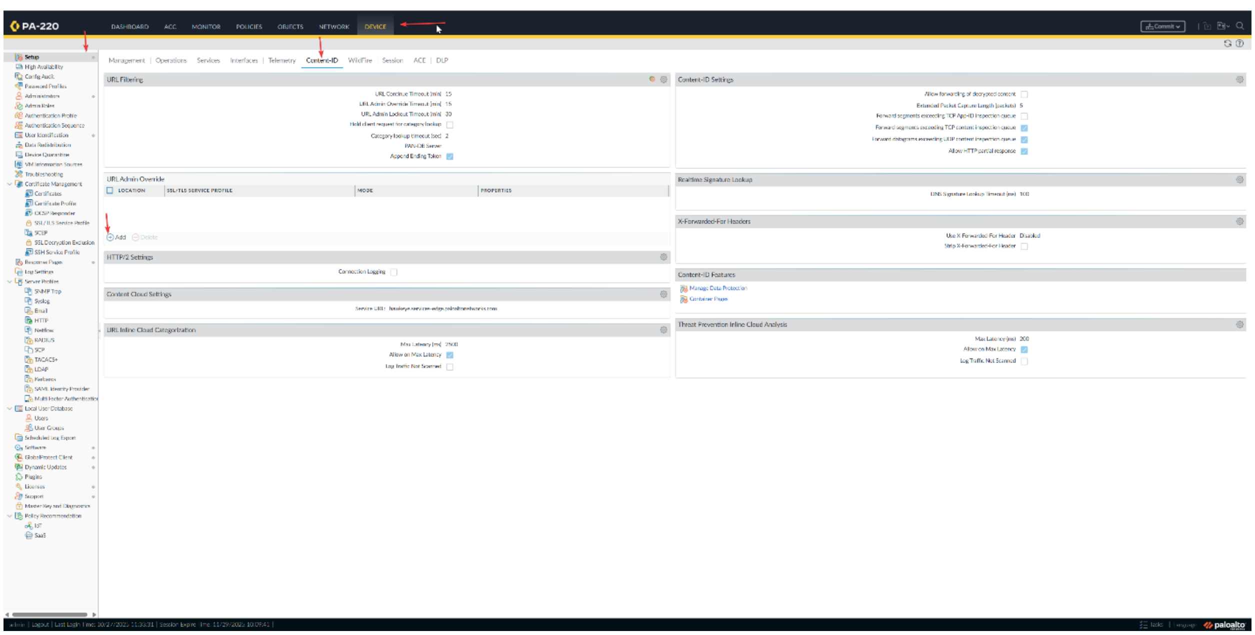
To apply the Url-filtering:
Click the “Actions” tab, set “Profile Type” to “Profiles”, and set the URL Filtering profile to the profile you set up

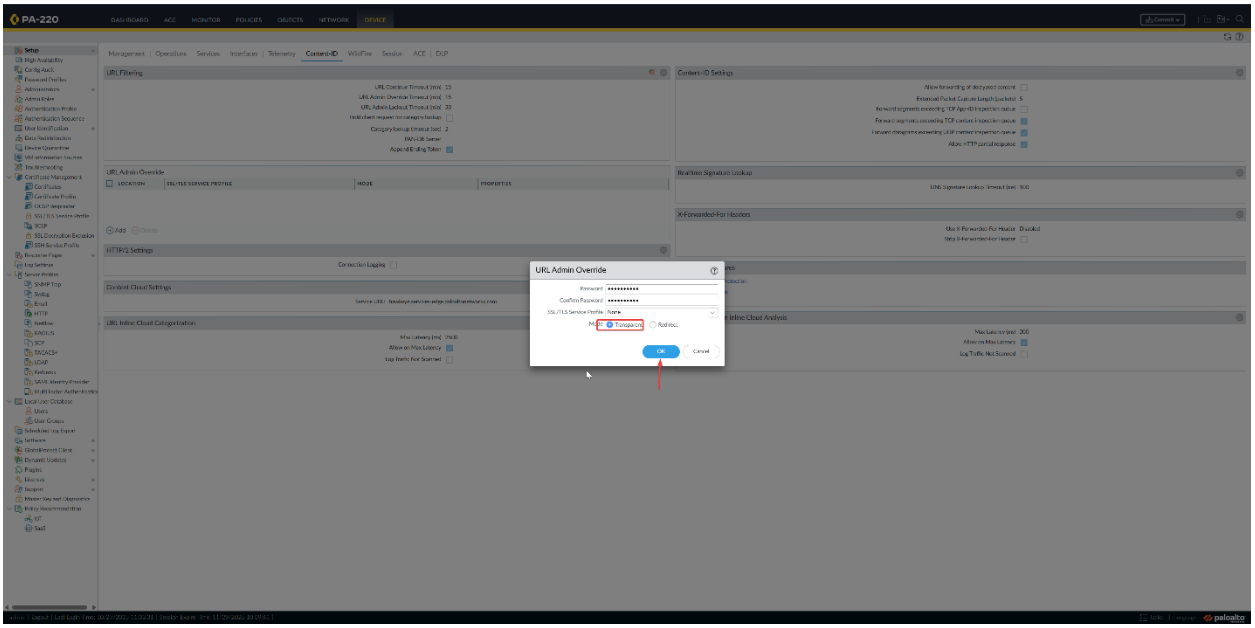




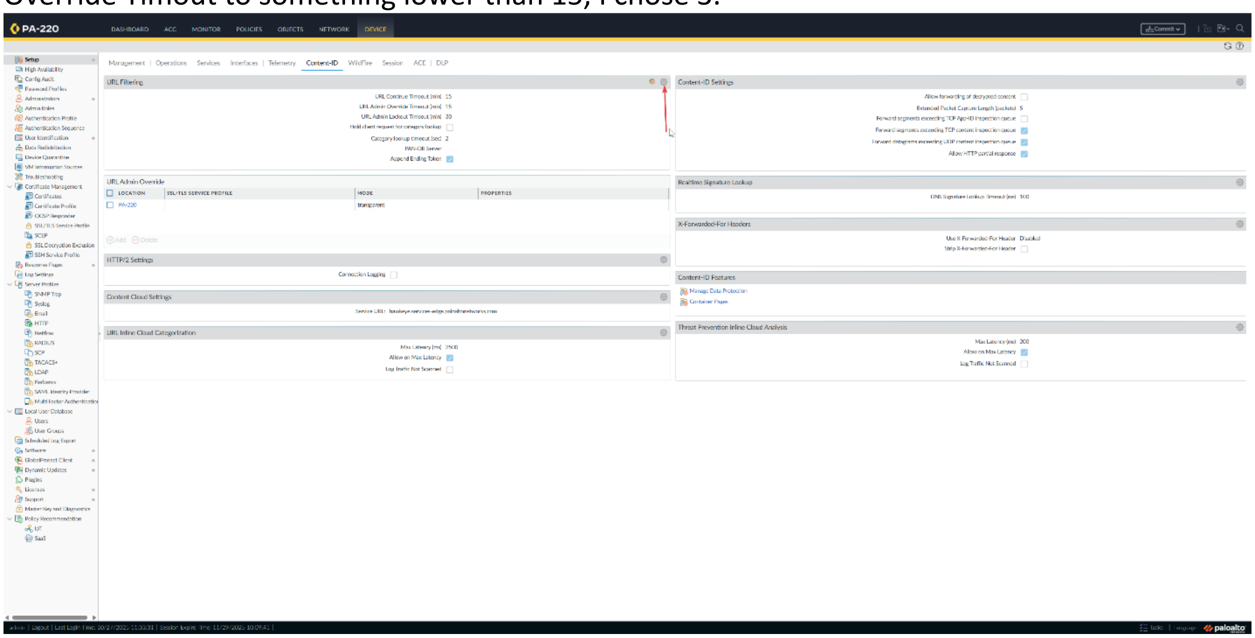
To configure Admin Override:

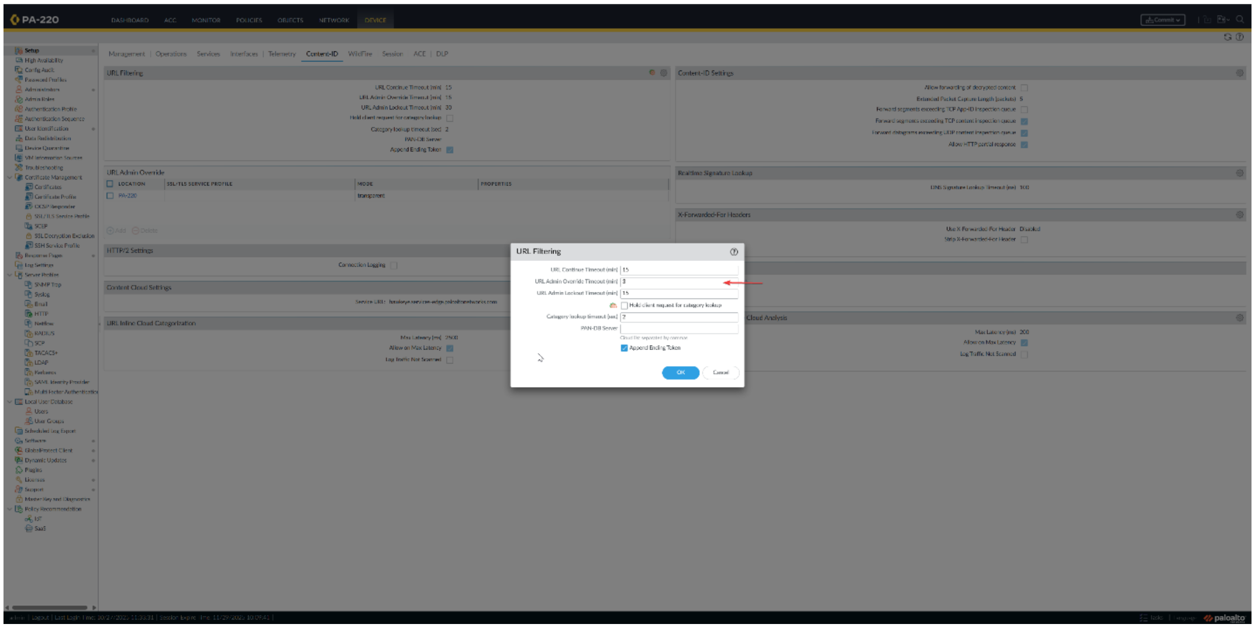
Go to “Device” > “Setup” > “Content ID” and click “Add” under the “URL Admin Override” section. Add your desired password, select transparent as your mode and click “OK”



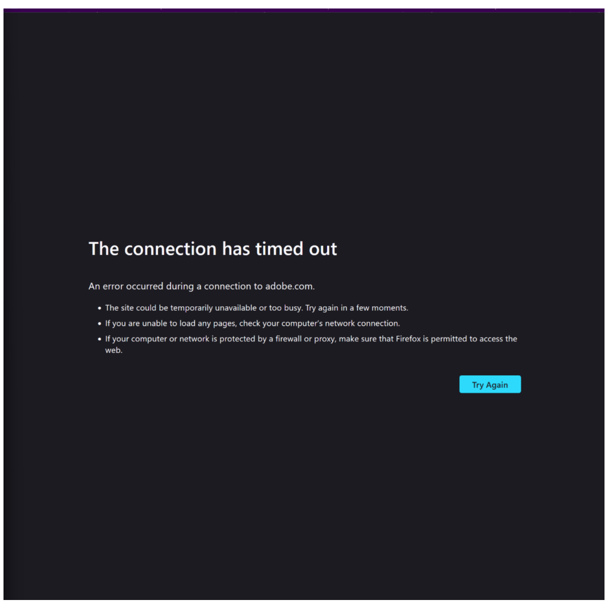


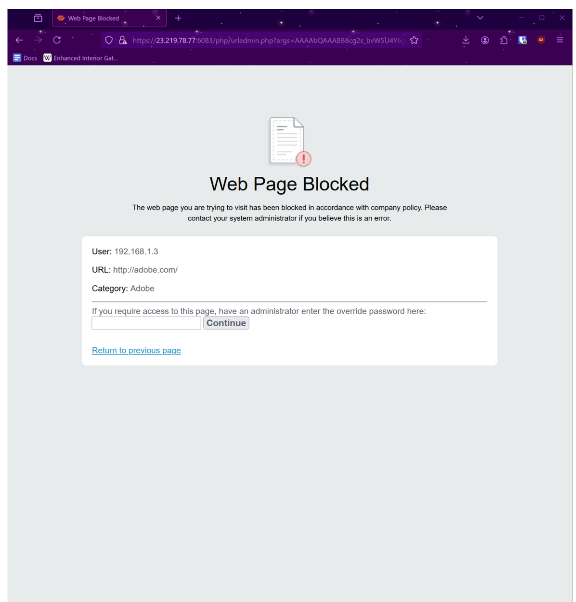
In the device tap click the settings icon on the URL filtering box and change the URL Admin Override Timout to something lower than 15, I chose 3.





Doing that should work and when you search up any websites in the categories in which you blocked or overridden, it should look like this:





Problems

Originally when we configured the URL filtering security policy to override for the category “Shopping” and configured the URL admin override, when we searched up amazon on the google browser, it continued to block the website. While troubleshooting, we tried to create the SSL/TLS profile and point it towards that in the Override, but it didn’t fix the issue, so we tried to use a different browser like Firefox, and it worked there.

Other problems that we went through were Initially when we blocked everything in our original URL filtering profile without testing and it didn't let us test on google, so we had to create a new url filtering object to just block a few categories to test.

Conclusion

To wrap up, the Palo Alto PA220 is a very capable device for small school environment. Through the PA220 web interface, setting up URL filtering is relatively straightforward. The application for these technical skills is endless. A lot of companies earn this level of cyber security in their machines. I hope I will use these skills in a company later at my job.